Assessment 2 Information

Research Project

The group research project is worth 25% of the final grade and is due at 11:59 pm on Friday 9 October 2026. Each project must be completed in a group of exactly two students, unless otherwise approved by the unit coordinator.

A central part of CITS4403 is the computational model your group will design, implement, explore, and analyse. The project will be assessed through the report, submitted code, and project demonstration. These components should complement one another: the report should explain the investigation, the code should reproduce the reported results, and the demonstration should present the model and key findings.

This assessment evaluates your ability to:

- formulate a focused research question about a complex system;
- select and justify an appropriate modelling framework;
- translate modelling assumptions into a computational implementation;
- design systematic experiments;
- analyse model behaviour using qualitative and quantitative evidence;
- interpret results critically and identify limitations; and
- communicate and collaborate effectively.

Project Structure

The choice of complex system is open-ended, but your project should demonstrate a clear connection to the concepts, behaviours, and computational modelling approaches covered in the unit.

Building a simulation alone is not sufficient. Your model must be used to investigate a clearly defined research question or hypothesis.

For example, you might examine how system behaviour changes with different parameters, interaction rules, network structures, interventions, or modelling assumptions.

A complete project should include:

1. a clearly defined system and motivation;
2. a focused research question or hypothesis;
3. a computational model with explicit assumptions and rules;
4. a justified experimental design;

5. qualitative and quantitative analysis;
6. interpretation of the results; and
7. discussion of limitations and possible extensions.

This is a fourth-year research project, and stronger projects will go beyond simply implementing an existing model. They should demonstrate depth in implementation and analysis, and include an element of independent investigation, such as exploring a new research question, extending an existing model, comparing alternative modelling assumptions, or investigating behaviour that has not been examined in the same way.

Start with a simple model, build your understanding of the system, and introduce additional complexity where it is justified. Use the lab sessions to discuss your ideas, modelling decisions, experiments, and results with the facilitators.

Project Checkpoints

The project includes three checkpoints. **Attendance at your group's scheduled checkpoint is compulsory.**

- **Checkpoint 1 —** Weeks 7–8: Project idea and research question
  - Discuss your proposed system, research question, modelling approach, and overall project direction with your facilitator.
  - You should also demonstrate that your group has set up a shared GitHub repository for project collaboration.
- **Checkpoint 2 —** Weeks 9–10: Model development and initial progress
  - Demonstrate the main components of your model and discuss your experimental plan, progress, and any issues you have encountered.
- **Checkpoint 3 —** Week 12: Project Demonstration
  - Demonstrate your completed model, present your main findings, and answer questions about your implementation, experiments, and analysis.

For Checkpoints 1 and 2, each group will have a 10-minute scheduled meeting with their facilitator. See your scheduled checkpoint time here (https://uniwa-my.sharepoint.com/:x:/r/personal/00105302_uwa_edu_au/Documents/Computational%20Modelling/Research%20Project/group_allocation.xlsx?d=w711996e857d242b6a1bb53af22e2e5ac&csf=1&web=1&e=s54hbk).

There will be **no compulsory checkpoint in Week 11**. The lab will instead be available as an optional drop-in consultation for groups who would like feedback, clarification, or help with their project. You may also use the dedicated Teams channel throughout the project for any group-project-related questions or discussion.

Further details on the technical requirements, report format, marking rubric, and submission instructions, including GitHub requirements, will be released shortly.